01 / PROBLEM AND CONTRIBUTION

GCN accelerator

EEE 525 · VLSI Design | Spring 2026 | Academic design

RESULT AT A GLANCE

Reproduced reference classifications at three implementation stages-RTL, post-synthesis, and post-route-using the same testbench through the ASIC flow.

The engineering question

How can graph-node classification be mapped into a controlled RTL datapath and carried through an ASIC design flow?

Situation

Graph-node classification combines dense feature transformation, sparse neighbor aggregation, and class selection. The project maps those phases into RTL and follows the design through physical implementation.

Task

Coordinate the datapath with an FSM, reproduce the reference classifications, and carry the design through synthesis and place-and-route.

My contribution

GCN datapath and controller design, simulation, synthesis, and physical implementation documented in the lab report.

Tools and concepts

Verilog / RTL / FSM design / Design Compiler / Cadence Innovus / Static timing analysis

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation step

Separated control and computation with a central FSM for feature loading, weight loading, transformation, aggregation, and classification.

02 / Implementation step

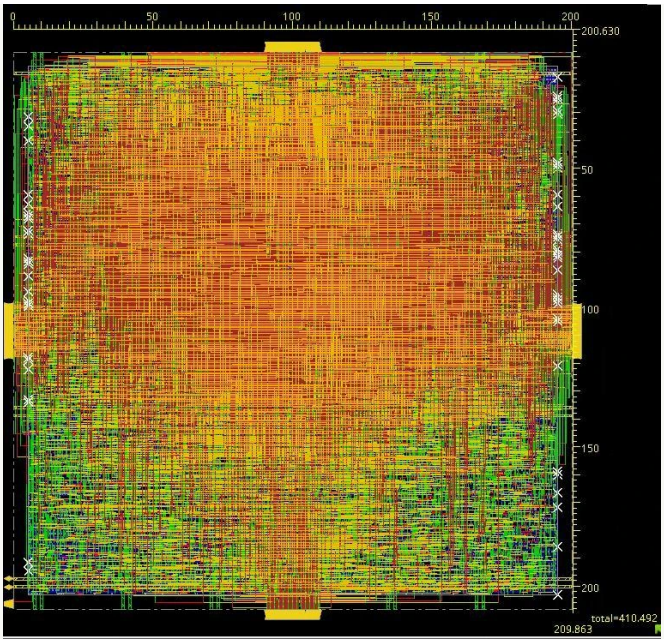
Used local weight storage and intermediate matrix buffers. The transformation stage computes feature-weight dot products; the aggregation stage traverses COO-format graph edges.

03 / Implementation step

Selected the largest of three output features for each node, then carried the RTL through Design Compiler synthesis and Innovus placement and routing.

Dense and sparse stages

Feature transformation computes feature-weight dot products; aggregation follows the sparse graph edge representation. Local storage and intermediate buffers connect these stages under the controller rather than treating the algorithm as one undifferentiated operation.



Routed accelerator layout

Cadence Innovus implementation view from the GCN report.

03 / RESULTS AND ENGINEERING PERSPECTIVE

Evidence, interpretation, and next steps

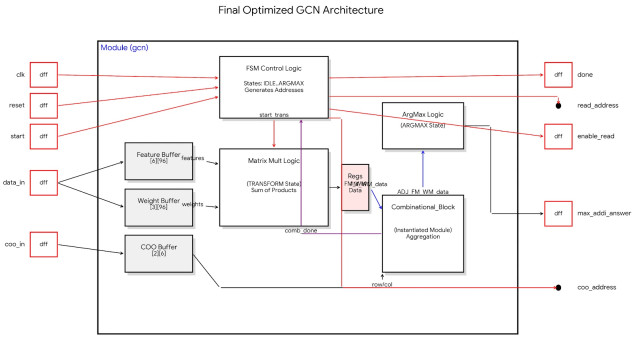
Validation stages	3
Class outputs	3

Reported result

The same testbench reproduced the reference classifications at RTL, post-synthesis, and post-route stages. Setup and hold timing were examined after routing.

Implementation continuity

The same reference-classification checks are used through RTL, synthesized, and routed design stages. This supports functional continuity for the tested data; examining timing reports is a separate implementation activity.



Datapath and controller

Reported GCN accelerator architecture.

Scope and limitations

Matching the reference classifications across implementation stages is the main result. The design also illustrates the separation of dense feature transformation from sparse graph aggregation.

Possible next steps - proposed, not completed results

- Add a reproducible input/reference-output example and tool invocation notes.
- Publish final timing constraints and unambiguous timing/area summaries before quoting additional PPA metrics.

Source material: EEE 525 Lab 4 milestone report, pp. 1-6, 8, 11-12; approved GCN brief.

Prepared from the supplied reports, project files, and verified portfolio copy. Figures retain their original content; explanatory diagrams are labelled.